Br	Br_2O	BrO_2	-	-	-	-
I	-	-	I_2O_5	-	-	I_4O_9 I_2O_4 (+4)

3.3.6 Oxoacids of halogens:

Chlorine forms four types of oxoacids namely hypochlorus acid, chlorous acid, chloric acid and perchloric acid. Bromine and iodine forms the similar acids except halous acid. However, flurine only forms hypofluorous acid. The oxidizing power oxo acids follows the order:

$$\mathbf{HOX > HXO_2 > HXO_3 > HXO_4}$$

Table 3.9

Type	HOX	HXO_2	HXO_3	HXO_4
	X = Cl, Br and I			
Common Name	Hypohalous acid	Halous acid	Halic acid	Perhalic acid
Oxidation state	+1	+3	+5	+7
F	HOF	-	-	-
Cl	HOCl	$HClO_2$	$HClO_3$	$HClO_4$
Br	HOBr		$HBrO_3$	$HBrO_4$
I	HOI		HIO_3	HIO_4

3.4 Group 18 (Inert gases) elements:

3.4.1 Occurrence:

All the noble gases occur in the atmosphere.

Physical properties:

As we move along the noble gas elements, their atomic radius and boiling point increases from helium to radon. The first ionization energy decreases from helium to radon. Noble gases have the largest ionisation energy compared to any other elements in a given row as they have completely filled orbital in their outer most shell. They are extremely stable and have a small tendency to gain or lose electrons. The common physical properties of the group 18 elements are listed in the Table.

Property	Neon	Argon	Krypton	Xenon	Radon
Physical state at 293 K	Gas	Gas	Gas	Gas	Gas
Atomic Number	10	18	36	54	86

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